

Assignment-1

04-07-2026: Normalization

Normalization:

1. Normalize the following table into 1 Normal form:

Member_Id	First_Name	Last_Name	Hobbies
101	Jayson	Mark	Cricket,Swimming,Football
102	Ram	Ganesh	Swimming,Running,Music
103	Raj	Kishore	Dancing, Singing, Running

2. Normalize the following Table into 2nd Normal Form:

Empno	Training	Training_Date	Dept
101	Oracle SQL	12-Aug-2015	IT
101	Java	21-Aug-2015	BU
102	Oracle SQL	18-Sep-2014	IT

3. Normalize the following into 3rd Normal Form:

Member_Id	First_Name	Last_Name	Sports	Fees
101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Foodball	90

1NF:

Member_Id	First_Name	Last_Name	Hobby
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

2NF:

Composite key=(Empno, Training)

Employee Training Table:

Empno	Training	Training Date
101	Oracle SQL	12-Aug-2015
101	Java	12-Aug-2015
102	Oracle SQL	18-Sep-2014

Empno	Dept
101	IT
102	IT

3NF:

Functional Dependencies:

Member_id --> First_Name,Last_Name,Sports
Sports --> Fees

Member Table:

Member_Id	First_Name	Last_Name	Sports
101	Rajesh	Chand	Cricket
102	Jayesh	Raj	Hockey
103	Mark	Dorson	Football

Sports_Fee Table:

Sports	Fees
Cricket	100
Hockey	80
Football	90

Assignment-2

07-07-2026: Retriving data using the sql select statement

1. Determine the structure of DEPT table and its content.

```
SQL> create table dept(
  2  dno number(2) primary key,
  3  dname varchar2(14),
  4  loc varchar2(13));

Table created.

SQL> insert into dept values(10,'Accounting','New york');
1 row created.

SQL> insert into dept values(20,'Research','Usa');
1 row created.

SQL> insert into dept values(30,'Sales','India');
1 row created.

SQL> commit;
Commit complete.

SQL> desc dept;
Name Null? Type
-----
DNO NOT NULL NUMBER(2)
DNAME VARCHAR2(14)
LOC VARCHAR2(13)

SQL> select * from dept;

DNO DNAME LOC
-----
10 Accounting New york
20 Research Usa
30 Sales India
```

2. Determine the structure of Emp table and its content.

```
SQL> create table employee(
  2  eno number(4) primary key,
  3  ename varchar2(20),
  4  deptNo number(2));

Table created.

SQL> INSERT INTO EMPLOYEE VALUES (7369,'SMITH',20);
1 row created.

SQL> INSERT INTO EMPLOYEE VALUES (7499,'BOB',30);
1 row created.

SQL> INSERT INTO EMPLOYEE VALUES (7788,'JOHN',20);
1 row created.

SQL> commit;
Commit complete.

SQL> desc employee;
Name Null? Type
-----
ENO NOT NULL NUMBER(4)
ENAME VARCHAR2(20)
DEPTNO NUMBER(2)

SQL> select * from employee;

ENO ENAME DEPTNO
-----
7369 SMITH 20
7499 BOB 30
7788 JOHN 20
```

3. Display the Ename and DeptNo from Emp table whose EmpNo is 7788.

```
SQL> select ename,deptno from employee where eno = 7788;

ENAME                DEPTNO
-----
JOHN                  20

SQL> |
```

Assignment-3

08-07-2026: Restricting and sorting Data

1	M	Display the Ename,Sal,Comm from Emp table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause
2	M	The HR department needs a query to display all unique job codes from the EMP table.
3	M	The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp #, Employee, Job, and Hire Date, respectively by giving Column Alias.
4	M	The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the job ID (separated by a comma and space) and name the column Employee and Title by giving Column Alias.
5	M	To familiarize yourself with the data in the EMP table, create a query to display all the data from that table. Separate each column output by a comma. Ename,Job,HIREDATE,MGR.Name the column title THE_OUTPUT.
6	M	Create a report to display the ename, job , and Hiredate for the employees name is SCOTT or TURNER. Order the query in ascending order by hiredate.
7	M	Display the ename and department number of all employees in departments 20 or 30 in ascending alphabetical order by ename.
8	M	Modify the previous query to display the last name and salary of employees who earn between 2000 and 3000 and are in department 20 or 30. Label the columns Employee and Monthly Salary, respectively giving Column Alias.
9	M	The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981
10	M	Display the Ename, Sal of employees who earn more than an amount the user specifies after a Prompt.
11	M	Create a report to display the last name and job title of all employees who do not have a manager.
12	M	Create a query that prompts the user for Manager ID and generate EMPNO,ENAME, SAL,DEPTNO. The user should have the ability to sort the records on a selected Column.
13	M	The HR department wants to run reports based on a manager. Create a query that prompts the user for a MGR and generates the empno, ename, salary, and department for that manager's employees. The HR department wants the ability to sort the report on a selected column
14	M	Display all employee last names in which the third letter of the name is A
15	M	Display the last name of all employees who have both an A and an S in their ename
16	M	Display the Ename, Job, Sal for all employees whose jobs are CLERK and whose salary is in 800 or 950 or 1300.

Solutions:

```

SQL> create table emp08(
2   eno int,
3   ename varchar(20),
4   job varchar(20)
5   mgr int,
6   hiredate date,
7   sal int,
8   comm int,
9   deptno int);
mgr int,
*
ERROR at line 5:
ORA-00907: missing right parenthesis

SQL> create table emp08(
2   eno int,
3   ename varchar(20),
4   job varchar(20),
5   mgr int,
6   hiredate date,
7   sal int,
8   comm int,
9   deptno int);

Table created.

SQL> desc emp08;
Name                                         Null?    Type
-----
ENO                                           NUMBER(38)
ENAME                                         VARCHAR2(20)
JOB                                           VARCHAR2(20)
MGR                                           NUMBER(38)
HIREDATE                                       DATE
SAL                                           NUMBER(38)
COMM                                           NUMBER(38)
DEPTNO                                        NUMBER(38)

SQL> insert into emp08 values(7788,'SCOTT','ANALYST',7566,TO_DATE('1987-04-19','YYYY-MM-DD'),3000,NULL,20);
1 row created.

SQL> insert into emp08 values(7844,'TURNER','SALESMAN',7698,TO_DATE('1981-09-08','YYYY-MM-DD'),1500,0,30);
1 row created.

SQL> insert into emp08 values(7839,'KING','PRESIDENT',NULL,TO_DATE('1981-11-17','YYYY-MM-DD'),5000,NULL,10);
1 row created.

SQL> insert into emp08 values(7900,'JAMES','CLERK',7698,TO_DATE('1981-12-03','YYYY-MM-DD'),950,NULL,30);
1 row created.

SQL> insert into emp08 values(7934,'MILLER','CLERK',7782,TO_DATE('1982-01-23','YYYY-MM-DD'),1300,NULL,10);
1 row created.

SQL> insert into emp08 values(7566,'JONES','MANAGER',7839,TO_DATE('1981-04-02','YYYY-MM-DD'),2975,NULL,20);
1 row created.

SQL> select ename,sal,comm from emp08 where comm IS NOT NULL order by 2 desc, 3 desc;
ENAME          SAL      COMM
-----
TURNER         1500      0

SQL> select DISTINCT job from emp08;
JOB
-----
SALESMAN
CLERK
PRESIDENT
MANAGER
ANALYST

SQL> select ename || ' , ' || job as "Employee and Title" from emp08;
Employee and Title
-----
SCOTT , ANALYST
TURNER , SALESMAN
KING , PRESIDENT
JAMES , CLERK
MILLER , CLERK
JONES , MANAGER

6 rows selected.

SQL> select eno as "Emp #",ename as "Employee", job as "Job", hiredate as "Hire Date" from emp08;
Emp # Employee          Job          Hire Date
-----
7788 SCOTT                ANALYST      19-APR-87
7844 TURNER              SALESMAN     08-SEP-81
7839 KING                PRESIDENT    17-NOV-81
7900 JAMES                CLERK        03-DEC-81
7934 MILLER              CLERK        23-JAN-82
7566 JONES                MANAGER      02-APR-81

6 rows selected.

```

```

SQL> select ename ||' , '||job||' , '|| hiredate||' , '||mgr as THE_OUTPUT from emp08;

THE_OUTPUT
-----
SCOTT , ANALYST , 19-APR-87 , 7566
TURNER , SALESMAN , 08-SEP-81 , 7698
KING , PRESIDENT , 17-NOV-81 ,
JAMES , CLERK , 03-DEC-81 , 7698
MILLER , CLERK , 23-JAN-82 , 7782
JONES , MANAGER , 02-APR-81 , 7839

6 rows selected.

SQL> select ename, job,hiredate from emp08 where ename IN('SCOTT','TURNER') order by giredate;
select ename, job,hiredate from emp08 where ename IN('SCOTT','TURNER') order by giredate
*
ERROR at line 1:
ORA-00904: "GIREDATE": invalid identifier

SQL> select ename, job,hiredate from emp08 where ename IN('SCOTT','TURNER') order by hiredate;

ENAME          JOB          HIREDATE
-----
TURNER          SALESMAN      08-SEP-81
SCOTT           ANALYST       19-APR-87

SQL> select ename, deptno from emp08 where deptno IN(20,30) order by ename;

ENAME          DEPTNO
-----
JAMES          30
JONES          20
SCOTT          20
TURNER         30

SQL> select ename as Employee, sal as "Monthly Salary" from emp08 where sal BETWEEN 2000 AND 3000 AND TO_CHAR(hiredate,'YYYY')=1981;

EMPLOYEE      Monthly Salary
-----
JONES          2975

SQL> select ename, sal from emp08 where sal>&salary;
Enter value for salary: 60000
old 1: select ename, sal from emp08 where sal>&salary
new 1: select ename, sal from emp08 where sal>60000

no rows selected

SQL> select ename,job from emp08 where mgr IS NULL;

ENAME          JOB
-----
KING           PRESIDENT

SQL> select eno,ename,sal,deptno from emp08 where mgr = &mgrid order by &column;
Enter value for mgrid: 7698
Enter value for column: 2
old 1: select eno,ename,sal,deptno from emp08 where mgr = &mgrid order by &column
new 1: select eno,ename,sal,deptno from emp08 where mgr = 7698 order by 2

      ENO ENAME          SAL      DEPTNO
-----
      7900 JAMES          950       30
      7844 TURNER        1500      30

SQL> select eno,ename,sal,deptno from emp08 where mgr = &mgrid order by &column;
Enter value for mgrid: 7566
Enter value for column: 'sal'
old 1: select eno,ename,sal,deptno from emp08 where mgr = &mgrid order by &column
new 1: select eno,ename,sal,deptno from emp08 where mgr = 7566 order by 'sal'

      ENO ENAME          SAL      DEPTNO
-----
      7788 SCOTT          3000      20

```

```

SQL> select ename from emp08 where ename like '%A%' AND ename like '%S%';

ENAME
-----
JAMES

SQL> select ename from emp08 where ename like '_A%';

ENAME
-----
JAMES

SQL> select ename,job,sal from emp08 where job='CLERK' AND sal IN (800,950,1300);

ENAME          JOB          SAL
-----
JAMES          CLERK          950
MILLER         CLERK          1300

SQL> |

```

Assignment-4

09-07-2026: Using single row functions to customize the output.

1	M	Write a query to display the current date. Label the column Date.
2	M	The HR department needs a report to display the employee number, last name, salary, and salary increased by 15.5% (expressed as a whole number) for each employee. Label the column New Salary.
3	M	Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column Increase
4	M	Write a query that displays the ename (with the first letter uppercase and all other letters lowercase) and the length of the ename for all employees whose name starts with the letters J, A, or M. Give each column an appropriate label. Sort the results by the employees' enames.
5	M	Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters H when prompted for a letter, then the output should show all employees whose last name starts with the letter H.
6	M	The HR department wants to find the length of employment for each employee. For each employee, display the ename and calculate the number of months between today and the date on which the employee was hired. Label the column MONTHS_WORKED. Order your results by the number of months employed. Round the number of months up to the closest whole number.
7	M	Create a report that produces the following for each employee: earns monthly but wants <3 times salary>. Label the column Dream Salaries.
8	M	Create a query to display the last name and salary for all employees. Format the salary to be 15 characters long, left-padded with the \$ symbol. Label the column SALARY.
9	M	Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column REVIEW. Format the dates to appear in the format similar to "Monday, the Thirty-First of July, 2000."
10	M	Display the last name, hire date, and day of the week on which the employee started. Label the column DAY. Order the results by the day of the week, starting with Monday.
11	M	Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show "No Commission." Label the column COMM.
12	M	Create a query that displays the first eight characters of the employees' last names and indicates the amounts of their salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column EMPLOYEES_AND_THEIR_SALARIES.
13	M	Using the DECODE function, write a query that displays the grade of all employees based on the value of the column JOB_ID, using the following data: PRESIDENT-A,MANAGER-B,SALESMAN-C,CLERK-D
14	M	Rewrite the statement in the preceding exercise using the CASE syntax

```
SQL> SELECT INITCAP(ENAME) AS NAME,
2      LENGTH(ENAME) AS LENGTH
3      FROM EMP08
4      WHERE ENAME LIKE 'J%'
5            OR ENAME LIKE 'A%'
6            OR ENAME LIKE 'M%'
7      ORDER BY ENAME;
```

NAME	LENGTH
Allen	5
Jones	5
Miller	6

```
SQL> desc emp08;
Name                                         Null?      Type
-----
ENO                                           NUMBER(38)
ENAME                                         VARCHAR2(20)
JOB                                           VARCHAR2(20)
MGR                                           NUMBER(38)
HIREDATE                                       DATE
SAL                                           NUMBER(38)
COMM                                          NUMBER(38)
DEPTNO                                        NUMBER(38)

SQL> insert into emp08 values(7369,'SMITH','CLERK',7902,DATE '1980-12-17',800,NULL,20);
1 row created.

SQL> insert into emp08 values(7499,'ALLEN','SALESMAN',7698,DATE '1981-02-20',1600,300,30);
1 row created.

SQL> insert into emp08 values(7566,'JONES','MANAGER',7839,DATE '1981-04-02',2975,NULL,20);
1 row created.

SQL> insert into emp08 values(7698,'BLAKE','MANAGER',7839,DATE '1981-05-01',2850,NULL,30);
1 row created.

SQL> insert into emp08 values(7788,'SCOTT','ANALYST',7566,DATE '1987-04-19',3000,NULL,20);
1 row created.

SQL> insert into emp08 values(7839,'KING','PRESIDENT',NULL,DATE '1981-11-17',5000,NULL,10);
1 row created.

SQL> insert into emp08 values(7844,'TURNER','SALESMAN',7698,DATE '1981-09-08',1500,0,30);
1 row created.

SQL> insert into emp08 values(7934,'MILLER','CLERK',7782,DATE '1982-01-23',1300,NULL,10);
1 row created.

SQL> commit;
Commit complete.
```

```
SQL> select eno,ename,sal,sal*10155 as NEW_SALARY from emp08;

      ENO ENAME                SAL NEW_SALARY
-----
      7369 SMITH                800    8124000
      7499 ALLEN               1600   16248000
      7566 JONES               2975   30211125
      7698 BLAKE               2850   28941750
      7788 SCOTT               3000   30465000
      7839 KING                5000   50775000
      7844 TURNER              1500   15232500
      7934 MILLER              1300   13201500

8 rows selected.

SQL> select eno,ename,sal,sal*1.155 as NEW_SALARY, (sal*1.155)-sal as INCREASE from emp08;

      ENO ENAME                SAL NEW_SALARY  INCREASE
-----
      7369 SMITH                800         924        124
      7499 ALLEN               1600        1848         248
      7566 JONES               2975    3436.125    461.125
      7698 BLAKE               2850    3291.75    441.75
      7788 SCOTT               3000     3465         465
      7839 KING                5000     5775         775
      7844 TURNER              1500     1732.5     232.5
      7934 MILLER              1300     1501.5     201.5

8 rows selected.
```

```
SQL> select ename from emp08 where ename LIKE UPPER('&LETTER') || '%';
Enter value for letter: T
old 1: select ename from emp08 where ename LIKE UPPER('&LETTER') || '%'
new 1: select ename from emp08 where ename LIKE UPPER('T') || '%'

ENAME
-----
TURNER

SQL> select ename , round(MONTHS_BETWEEN(sysdate,hiredate)) as MONTHS_WORKED from emp08 order by MONTHS_WORKED

ENAME                MONTHS_WORKED
-----
SCOTT                 471
MILLER                534
KING                 536
TURNER                538
BLAKE                 542
JONES                 543
ALLEN                 545
SMITH                 547

8 rows selected.

SQL> select ename,sal,sal*3 as "Dream salaries" from emp08;

ENAME                SAL Dream salaries
-----
SMITH                 800         2400
ALLEN                1600         4800
JONES                2975         8925
BLAKE                2850         8550
SCOTT                3000         9000
KING                 5000        15000
TURNER               1500         4500
MILLER               1300         3900

8 rows selected.
```

```
SQL> select ename, LPAD('$' || sal,15,'$') as salary from emp08;

ENAME
-----
SALARY
-----
SMITH
$$$$$$$$$$$$$800

ALLEN
$$$$$$$$$$$$$1600

JONES
$$$$$$$$$$$$$2975

ENAME
-----
SALARY
-----
BLAKE
$$$$$$$$$$$$$2850

SCOTT
$$$$$$$$$$$$$3000

KING
$$$$$$$$$$$$$5000

ENAME
-----
SALARY
-----
TURNER
$$$$$$$$$$$$$1500

MILLER
$$$$$$$$$$$$$1300

8 rows selected.
```

```
SQL> select ename,hiredate,TO_CHAR(hiredate,'DAY') as DAY from emp08 order by TO_CHAR(hiredate,'D');
```

ENAME	HIREDATE	DAY
SCOTT	19-APR-87	SUNDAY
KING	17-NOV-81	TUESDAY
TURNER	08-SEP-81	TUESDAY
SMITH	17-DEC-80	WEDNESDAY
JONES	02-APR-81	THURSDAY
ALLEN	20-FEB-81	FRIDAY
BLAKE	01-MAY-81	FRIDAY
MILLER	23-JAN-82	SATURDAY

8 rows selected.

```
SQL> select ename, NVL(TO_CHAR(comm),'No commission') as comm from emp08;
```

ENAME	COMM
SMITH	No commission
ALLEN	300
JONES	No commission
BLAKE	No commission
SCOTT	No commission
KING	No commission
TURNER	0
MILLER	No commission

8 rows selected.

```
SQL> select substr(ename,1,8) as EMPLOYEES_AND_THEIR_SALARIES, rpad('*',trunc(sal/1000),'*') as SALARY from emp08 order by sal desc;
```

EMPLOYEES_AND_THEIR_SALARIES

SALARY

KING

SCOTT

JONES
**

EMPLOYEES_AND_THEIR_SALARIES

SALARY

BLAKE
**

ALLEN
*

TURNER
*

EMPLOYEES_AND_THEIR_SALARIES

SALARY

MILLER
*

SMITH

8 rows selected.


```
SQL> select ename,job,decode(job,'PRESIDENT','A','MANAGER','B','SALESMAN','C','CLERK','D','OTHERS') as GRADE from emp88;
```

ENAME	JOB	GRADE
SMITH	CLERK	D
ALLEN	SALESMAN	C
JONES	MANAGER	B
BLAKE	MANAGER	B
SCOTT	ANALYST	OTHERS
KING	PRESIDENT	A
TURNER	SALESMAN	C
MILLER	CLERK	D

8 rows selected.

```
SQL> select ename,job,CASE when job='PRESENT' then 'A'
2  when job ='MANAGER' then 'B'
3  when JOB='CLERK' then 'D'
4  when JOB='SALESMAN' THEN 'C'
5  else 'OTHER'
6  END as GRADE from emp88;
```

ENAME	JOB	GRADE
SMITH	CLERK	D
ALLEN	SALESMAN	C
JONES	MANAGER	B
BLAKE	MANAGER	B
SCOTT	ANALYST	OTHER
KING	PRESIDENT	OTHER
TURNER	SALESMAN	C
MILLER	CLERK	D

8 rows selected.

```
SQL> |
```

Assignment-5

10-07-2026 : Reporting Aggregated Data using the Group functions

1	M	Find the highest, lowest, sum, and average salary of all employees. Label the columns
2	M	Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.
3	M	Modify the above query to display the minimum, maximum, sum, and average salary for each job type.
4	M	Write a query to display the number of people with the same job
5	M	Determine the number of managers without listing them. Label the column Number of Managers
6	M	Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.
7	M	Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$2000 or less. Sort the output in descending order of salary.
8	M	Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.

```

SQL> select MAX(sal) as MAXIMUM, MIN(sal) as MINIMUM, SUM(sal) as SUM, ROUND(AVG(sal)) as AVERAGE from emp08;
-----
MAXIMUM      MINIMUM      SUM      AVERAGE
-----
5000          800      19025      2378

SQL> select job, min(sal) as MINIMUM, max(sal) as MAXIMUM, sum(sal) as SUM, round(avg(sal)) as AVERAGE from emp08 group by job;
-----
JOB              MINIMUM      MAXIMUM      SUM      AVERAGE
-----
CLERK              800          1300          2100      1050
SALESMAN           1500          1600          3100      1550
PRESIDENT          5000          5000          5000      5000
MANAGER            2850          2975          5825      2913
ANALYST            3000          3000          3000      3000

SQL> select job,count(*) as NUMBER_OF_PEOPLE from emp08 group by job;
-----
JOB              NUMBER_OF_PEOPLE
-----
CLERK              2
SALESMAN           2
PRESIDENT          1
MANAGER            2
ANALYST            1

SQL> select count(DISTINCT mgr) as "Number of managers" from emp08 where mgr IS NOT NULL;
-----
Number of managers
-----
5

SQL> select max(sal)-min(sal) as DIFFERENCE from emp08;
-----
DIFFERENCE
-----
4200

SQL> select mgr,min(sal) as SALARY from emp08
2 where mgr IS NOT NULL
3 group by mgr
4 having min(sal)>2000
5 order by SALARY DESC;
-----
MGR      SALARY
-----
7566      3000
7839      2850

SQL> SELECT COUNT(*) AS TOTAL_EMPLOYEES,
2 SUM(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1980' THEN 1 ELSE 0 END) AS HIRED_1980,
3 SUM(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1981' THEN 1 ELSE 0 END) AS HIRED_1981,
4 SUM(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1982' THEN 1 ELSE 0 END) AS HIRED_1982
5 FROM EMP08;

TOTAL_EMPLOYEES HIRED_1980 HIRED_1981 HIRED_1982
-----
8 1 5 1

SQL> |

```

Assignment-6

11-07-2026: Using the set Operators

No.	Cat	Hands-on Assignment
1	M	Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 10, 20, and 30, giving each column an appropriate heading.
2	M	Using set operator display the DEPTNO,SUM(SAL) for each dept, JOB,SUM(SAL) for each Job and Total Salary.
3	M	Using Set Operator display the JOB and Deptno in employees working in deptno 20,10,30 in that order.

```

SQL> select job,
2 sum(case when deptno=10 then sal else 0 end) as dept10,
3 sum(case when deptno=20 then sal else 0 end) as dept20,
4 sum(case when deptno=30 then sal else 0 end) as dept30,
5 sum(sal) as total
6 from emp1
7 group by job;
-----
JOB              DEPT10      DEPT20      DEPT30      TOTAL
-----
clerk              1300          800          950          3050
salesman           0              0          3100          3100
manager            0          2975          2850          5825
analyst            0          3000           0          3000
president          5000           0           0          5000

```

```

SQL> select to_char(deptno) as category,
2      sum(sal) as total_salary
3      from emp1
4      group by deptno
5      union
6      select job as category,
7      sum(sal) as total_salary
8      from emp1
9      group by job
10     union
11     select 'total' as category,
12     sum(sal) as total_salary
13     from emp1;

```

CATEGORY	TOTAL_SALARY
20	6775
30	6900
10	6300
clerk	3050
salesman	3100
manager	5825
analyst	3000
president	5000
total	19975

9 rows selected.

```

SQL> select job,deptno
2      from emp1
3      where deptno=20
4      union
5      select job,deptno
6      from emp1
7      where deptno=10
8      union
9      select job,deptno
10     from emp1
11     where deptno=30;

```

JOB	DEPTNO
clerk	20
manager	20
analyst	20
president	10
clerk	10
salesman	30
manager	30
clerk	30

8 rows selected.

Assignment-7

13-07-2026: Displaying data from multiple tables using joins:

1	M	Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME,SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.	Natural Join
2	M	The HR department needs a report of all employees. Write a query to display the JOB,MGR,SAL,COMM,DNAME of employees whose JOB is SALESMAN.	Equi Join
3	M	The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO, and DNAME for all employees who work in DALLAS.	Equi Join
4	M	Create a report to display employees' ename and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.	Self Join
5	M	Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.	Outer Join
6	M	The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.	Non-Equi Join
7	M	Display the ENAME,DNAME of all the employees. Also display those department name which do not have any employees working.	Outer Join
8	M	The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.	Self Join with Outer Join

9	M	Display the EMPNO,ENAME,DNAME,LOC of those employees who are working as CLERK. Use the USING clause.	USING Clause
10	M	Display the ENAME,SAL,MGR,DNAME of employees whose salary is more than 2000. Use the ON clause.	On Clause
11	M	Display the EMPNO,ENAME,JOB,DEPTNO,DNAME,LOC of employees. Use LEFT OUTER JOIN.	LEFT OUTER JOIN
12	M	Display the ENAME,DNAME of employees. Use RIGHT OUTER JOIN.	RIGHT OUTER JOIN
13	M	Display the EMPNO,DNAME,LOC of employees. Use FULL OUTER JOIN.	FULL OUTER JOIN

```
SQL> select empno,ename,sal,dname,location
2 from emp1
3 natural join dept;
```

EMPNO	ENAME	SAL	DNAME	LOCATION
7369	smith	800	RESEARCH	DALLAS
7499	allen	1600	SALES	CHICAGO
7566	jones	2975	RESEARCH	DALLAS
7698	blake	2850	SALES	CHICAGO
7788	scott	3000	RESEARCH	DALLAS
7839	king	5000	ACCOUNTING	NEW YORK
7844	turner	1500	SALES	CHICAGO
7900	james	950	SALES	CHICAGO
7934	millar	1300	ACCOUNTING	NEW YORK

9 rows selected.

```
SQL> select job,mgr,sal,comm,dname
2 from emp1 e,dept d
3 where e.deptno=d.deptno
4 and job='salesman';
```

JOB	MGR	SAL	COMM	DNAME
salesman	7698	1600	300	SALES
salesman	7698	1500	0	SALES

```
SQL> select ename,job,e.deptno,dname
2 from emp1 e,dept d
3 where e.deptno=d.deptno
4 and location='DALLAS';
```

ENAME	JOB	DEPTNO	DNAME
smith	clerk	20	RESEARCH
jones	manager	20	RESEARCH
scott	analyst	20	RESEARCH

```
SQL> select e.ename as employee,
2 e.empno as "emp#",
3 m.ename as manager,
4 m.empno as "mgr#",
5 from emp1 e,emp1 m
6 where e.mgr=m.empno;
```

EMPLOYEE	emp#	MANAGER	mgr#
scott	7788	jones	7566
allen	7499	blake	7698
turner	7844	blake	7698
james	7900	blake	7698
jones	7566	king	7839
blake	7698	king	7839

6 rows selected.

```
SQL> select e.ename as employee,
2 e.empno as "emp#",
3 m.ename as manager,
4 m.empno as "mgr#"
5 from emp1 e
6 left join emp1 m
7 on e.mgr=m.empno
8 order by e.empno;
```

EMPLOYEE	emp#	MANAGER	mgr#
smith	7369		
allen	7499	blake	7698
jones	7566	king	7839
blake	7698	king	7839
scott	7788	jones	7566
king	7839		
turner	7844	blake	7698
james	7900	blake	7698
millar	7934		

9 rows selected.

```
SQL> select ename,
2 job,
3 dname,
4 sal,
5 grade
6 from emp1 e,salgrade s,dept d
7 where e.deptno=d.deptno
8 and sal between losal and hisal;
```

ENAME	JOB	DNAME	SAL	GRADE	
millar	clerk	ACCOUNTING		1300	2
king	president	ACCOUNTING		5000	5
smith	clerk	RESEARCH	800	1	
jones	manager	RESEARCH	2975		4
scott	analyst	RESEARCH	3000		4
james	clerk	SALES	950	1	
allen	salesman	SALES	1600		3
turner	salesman	SALES	1500		3
blake	manager	SALES	2850		4

9 rows selected.

```
SQL> select ename,dname
2 from dept d
3 left join emp1 e
4 on d.deptno=e.deptno;
```

ENAME	DNAME
smith	RESEARCH
allen	SALES
jones	RESEARCH

```

blake      SALES
scott      RESEARCH
king       ACCOUNTING
turner     SALES
james      SALES
miller     ACCOUNTING
           OPERATIONS

```

10 rows selected.

```

SQL> select e.ename,
2         e.hiredate,
3         m.ename,
4         m.hiredate
5   from emp1 e
6  left join emp1 m
7    on e.mgr=m.empno
8  where e.hiredate<m.hiredate;

```

ENAME	HIREDATE	ENAME	HIREDATE
allen	20-FEB-81	blake	01-MAY-81
jones	02-APR-81	king	17-NOV-81
blake	01-MAY-81	king	17-NOV-81

```

SQL> select empno,ename,dname,location
2   from emp1
3  join dept
4    using(deptno)
5  where job='clerk';

```

EMPNO	ENAME	DNAME	LOCATION
7934	miller	ACCOUNTING	NEW YORK
7369	smith	RESEARCH	DALLAS
7900	james	SALES	CHICAGO

```

SQL> select ename,sal,mgr,dname
2   from emp1 e
3  join dept d
4    on e.deptno=d.deptno
5  where sal>2000;

```

ENAME	SAL	MGR	DNAME
king	5000		ACCOUNTING
jones	2975	7839	RESEARCH
scott	3000	7566	RESEARCH
blake	2850	7839	SALES

```

SQL> select empno,dname,location
2   from emp1 e
3  full outer join dept d
4    on e.deptno=d.deptno;

```

EMPNO	DNAME	LOCATION
7369	RESEARCH	DALLAS
7499	SALES	CHICAGO
7566	RESEARCH	DALLAS
7698	SALES	CHICAGO
7788	RESEARCH	DALLAS
7839	ACCOUNTING	NEW YORK
7844	SALES	CHICAGO
7900	SALES	CHICAGO
7934	ACCOUNTING	NEW YORK
	OPERATIONS	BOSTON

10 rows selected.